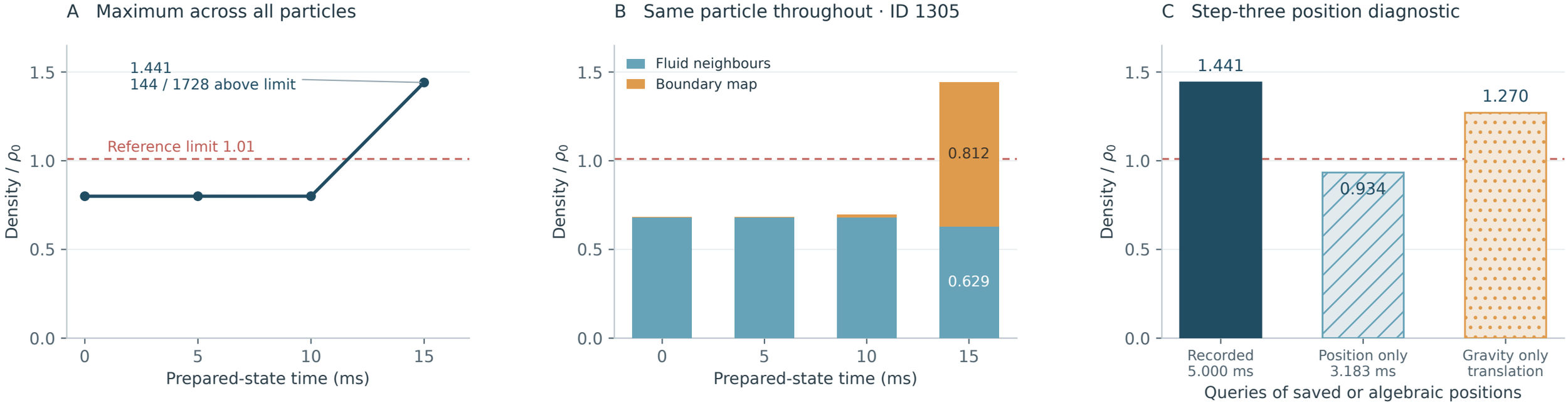
A source-consistent step can still exceed the density limit

1728-particle local cohort · archived states and boundary-map queries · no physical-flow acceptance



A: the maximizing particle may change at each time. B: density increase at the fixed particle is dominated by the boundary contribution.

C: the shorter-step query reuses the recorded velocity; forces and pressure are not re-solved. It diagnoses sensitivity, not a validated repair.

Hatched bars are not new simulated trajectories.